CM22009 – Machine Learning

Coursework 1: Regression Performance

Set: 19/10/2026 (Monday, Week 4) **Due:** 13/11/2026 (Friday, Week 7), 8pm
Percentage of overall unit mark: 15%
Submission Location: Moodle **Submission Components:** Poster **Submission Format:** 1 x PDF file and 1 x Jupyter Notebook
Anonymous Marking: Yes
Generative AI Assessment Categorisation: Open Lane [guidance]

1 Overview

This coursework assesses two of the three module Learning Outcomes:

- Demonstrate understanding of a wide range of machine learning techniques, their strengths and their limitations.
- Write code in a relevant programming language and employ software libraries to solve problems in machine learning.

For this coursework you will be selecting, implementing, and evaluating regression models for a scenario of your choice.

1.1 The Task

You must complete the following tasks, documenting your process and findings in the form of a poster (as defined in section 2):

- Select a scenario and research question from the options given in section 6.
- Train at least two different machine learning models to answer your selected research question.
- Evaluate your machine learning models using suitable criteria.
- Conclude by providing a recommended machine learning approach to the research question.

2 Deliverable

There are two deliverables to this coursework:

Deliverable	Format	Content
A3 Poster	.pdf file	Overview of your experiments, training process, evaluation and final recommendation.
Code	.ipynb	The code necessary to run your experiments, train the models and produce necessary graphics.

The poster must contain the following subsections:

Section	Expected Content
Introduction	Overview of the problem and poster: • What is this poster showing? • What is the primary research question?
Methodology	Details of the methodology employed: • What models are you chosen for the problem? • What pre-processing steps have you applied? • What data have you used and how have you split the data? • What hyperparameters have you selected? • Justify the choices made at each stage.
Results	Details of the results from training and evaluation: • What evaluation metrics are used and why? • Are the results clearly presented using suitable tables, figures? • How do the models compare in terms of performance? • Are the results interpreted and discussed, rather than simply reported?
Conclusions	Overall conclusions from the experiments and your recommendation: • What were the key findings and insights? • What were the limitations of your study? • What could you do differently to improve the study?

2.1 Mark Scheme

Your poster will be assessed using the contributions below, each section will be marked according to the marking grid provided at the end of this document. Whilst the code itself does not form part of the marking criteria it is required for evidence that the results are indeed true.

Criteria	Percentage of overall deliverable mark
Introduction	10%
Methodology	30%
Results	30%
Conclusions	20%
Presentation	10%

3 Feedback

Formative feedback on your work will be offered throughout the duration of the coursework:

- During your labs/tutorials, the tutors will be available to answer questions and offer guidance. Please note that tutors will not be able to make decisions on your behalf regarding your assignment. They are there to discuss your ideas and offer advice.
- Use Moodle forums to post general questions or questions specific to your assignment. The unit teaching team will respond to these, as well as your peers. This way, we will create a repository of knowledge that will be available to all.

You will receive **summative feedback** on your work within 15 working days of the submission deadline. The feedback will discuss your performance based on the criteria for marking, including what you did well and how specific components/sections could have been improved.

4 Academic Integrity

4.1 General Notes on Academic Integrity

Your work will be checked to ensure that you have not plagiarised. For more information about the plagiarism policy at the University, see: https://library.bath.ac.uk/referencing/plagiarism.

Any code submitted will also be checked.

4.2 Academic Integrity in the Open Lane

This coursework is **an Open Lane** assessment - Students have access to GenAI tools, and their use is permitted. Choosing not to use GenAI in the open lane is a legitimate choice. If you choose to use GenAI, the University's supported tool is **Copilot**. Ensure you are logged in using your Bath email address because this safeguards your user data.

<u>Please note</u>: When using GenAI, only upload material you hold the copyright to. Never upload confidential or commercially sensitive information — for example data given to you by a placement or industry partner — or a staff member's work without their permission.

Please also be aware of the following notes on responsibility, academic integrity, and resources.

4.2.1 Students remain responsible for the work they submit.

In the open lane, you remain fully responsible for your submitted work:

1. You are responsible for the content, the intellectual decisions, the judgement and the understanding contained in your submitted work.
2. You are accountable for any mistakes, omissions or inaccuracies, including any that GenAI introduced.
3. You must be able to explain, defend and critically evaluate every part of work that you submit. **The Department reserves the right to invite students to a viva or use alternative means of ensuring that work submitted as part of an assessment is the student's own intellectual product.**
4. Reference and acknowledge your use of GenAI following Library guidance and say where it shaped the work.

4.2.2 Academic integrity rules apply in full in the open lane.

Using GenAI in the open lane is not an offence in itself. It can, however, be the means for poor academic practice or academic misconduct.

1. **Submitting fabricated or unchecked sources** - Including references, quotations or data points produced by GenAI without verifying that they exist and say what you claim.
2. **Passing off GenAI output as your sole creation** - Presenting AI-generated text, ideas or analysis as though they were solely your creation.
3. **Substituting for your thinking** - Using AI to replace your intellectual contribution rather than to support it.

These may be investigated as plagiarism, fabrication, or collusion under Regulation 19 (Academic Misconduct), which also sets out the penalties that can apply.

4.2.3 Important resources for students on GenAI

How can GenAI support my learning?

Explore the Skills Centre workflow pages, including information on supported tools: https://unihub.bath.ac.uk/s/myskills/Workflows/Detail/264

How do I acknowledge GenAI use?

Follow the Library referencing guidance: https://library.bath.ac.uk/referencing/plagiarism

4.3 Using AI in This Assignment

This coursework is classified as ***Open Lane*** for specific defined processes within the assessment and its use is not mandatory in order to complete the assessment.

Allowed Uses of Generative AI Tools	Not Allowed Uses of Generative AI Tools
Brainstorming ideas (e.g., exploring scenarios or potential model choices)	Generating large sections of the final poster verbatim
Explaining concepts (e.g., regression, evaluation metrics, model pros/cons)	Fabricating results or analysis (e.g., using AI to make up performance metrics or conclusions without running experiments).
Assisting with code debugging or optimisation (must be reviewed and tested by you)	Submitting AI-generated code without understanding or testing it
Improve writing (e.g., grammar, clarity, style)	Presenting AI-generated analysis or discussion as your own original reasoning
Including a brief acknowledgement of GenAI use	Failing to acknowledge significant AI assistance where used.

If you use GenAI meaningfully (beyond simple grammar checks), briefly state this in your poster's acknowledgements, e.g.:

"*I used Copilot to help debug Python code and to rephrase some sentences for clarity. All results and conclusions are based on my own work and verification*."

5 FAQ

5.1 Is the quality of my models marked?

In short, no. We are more interested in whether you can adequately explain and evaluate your models' performances using the metrics and evaluation methods you choose. There are no direct marks available for the quality of the model.

5.2 Why do I only have one side of A3?

When faced with these challenges in the real world it is important that you can provide succinct but detailed reports to clients. There is no room for flamboyant language or unnecessary detail, I suggest you are creative in how you use the size limit with figures and tables where necessary, the less figures you need to use, the more space you have but sometimes a picture can paint a thousand words.

5.3 Do we have to write the algorithms ourselves?

No! The point of this exercise is to allow you the freedom to select and implement necessary methods from Python libraries, whilst you may wish to write the algorithms yourself this would have not fluency on your mark. Consider the task, the short time frame and remember this is primarily about your knowledge of evaluating and interpreting machine learning models using appropriate techniques, not how well you can program in Python.

5.4 How many methods do we need to evaluate?

To make a comparison you'll need at least 2 and there is no upper limit. However, think carefully again about the space given to report your results and the time you need to perform the evaluation.

5.5 Can we use methods that haven't been taught?

If you wish to use methods or validation metrics which you have found independently or used last semester you may use these if you make the adequate justifications in the poster. This will not attract more marks.

5.6 Does my poster have to be portrait or landscape?

You are free to use either portrait or landscape as long as it is A3.

5.7 Do you have any advice for creating a poster?

Posters should be eye-catching, straight to the point and graphical. For formal advice, the university has a blog on good poster tips: https://blogs.bath.ac.uk/technicians/2019/05/15/poster-writing-hints-and-tips/. Here is another useful resource: https://www.animateyour.science/post/how-to-write-engaging-headings-to-make-your-scientific-poster-pop.

5.8 How do I create an A3 poster?

The easiest way to create an A3 poster (and export to PDF) is by using Microsoft PowerPoint:

1. Open an empty PowerPoint with the starting slide.
2. Go to the 'Design' tab.
3. Click on 'Slide Size' (usually on the right-hand side of the toolbar).
4. Click 'Custom Slide Size…'.
5. Select A3 Paper.

Once you've finished your poster you can use the export function in the 'File' menu to save as PDF.

6 Scenarios

For this coursework you may choose from the following three scenarios:

Scenario	**Rainfall Forecasting**	**Predicting Forest Fire Extent**	**House Price Prediction**
Research Question	Can we predict monthly rainfall given forecasted minimum and maximum temperatures?	Can we predict the extent of a forest fire given meteorological conditions?	Can we predict house prices based on given features?
Description	Given the increasing threat to society caused by extreme weather events a new investigation has been launched into improving monthly weather forecasts. Forecasting the minimum and maximum temperatures of a given month is already possible thanks to mathematical models; however, rainfall forecasts are severely lacking and many hope machine learning could be the solution. You have been brought in to carry out a series of experiments using machine learning to forecast rainfall given the minimum and maximum temperature forecasts. Historic data for three weather stations in Wales have been provided for you.	Forest fires are becoming increasingly common in southern Europe and the extent of these forest fires can have drastic consequences on local societies and economies. Being able to predict the potential extent of forest fires can be crucial to enabling emergency services to take appropriate measures. You have been given a dataset of historic forest fires for a park in Portugal (Montesinho park) and have been asked to develop a regression model for predicting the extent of forest fires in the park given the range of features provided to you.	Estimating the price of a house prior to putting it on the market can be a critical task for estate agents to enable expectation management of their customers. You have been provided a dataset with three key features to house prices in central London along with the final selling price of the house on a per metre squared basis. You have been asked to develop a model for estimating the sale price (on a per metre squared basis) of a house given the features provided in the dataset.

Dataset	The dataset provided is in the form of three Numpy files (one for each weather station): • Cardiff.npy • Aberporth.npy • Valley.npy Each dataset contains 6 columns as follows: • Year: Year of reading. • Month: Month of reading. • Minimum Temperature: Minimum temperature forecast for the month in Celsius. • Maximum Temperature: Maximum temperature forecast for the month in Celsius. • No. Frost Days: The number of days forecast for given the month. • Monthly Rainfall: The amount of rainfall which fell in the month in mm. Each row represents one month. To load the dataset using numpy you can use the following command: dataset = np.load("<FILENAME>")	The dataset provided to you is in a single CSV file: • montesinho.csv The dataset contains 6 columns: • DMC Index: A numerical rating based on the moisture content of compact organic matter. • Temperature: Temperature of the park in Celsius. • Relative Humidity: Relative humidity as a %. • Wind: Wind speed in km/h. • Rainfall: Total rainfall in mm/m^2 • Area: The total area of the forest which burned (in hectares). Each row represents one reading. To load the dataset using numpy you can use the following command: dataset = np.loadtxt("montesinho.csv", delimiter=',')	This dataset is provided to you in a single CSV file: • realestate.csv The dataset contains 4 columns: • House Age: Years since the house was built. • Distance to Tube: The distance to the closest tube station in metres. • Nearby Stores: The number of stores within a 2km radius. • Price per m^2: the price the house sold for per m^2. Each row represents one previous house sale. To load the dataset using numpy you can use the following command: dataset = np.loadtxt("realestate.csv", delimiter=',')

Marking Rubric

Section	%	Mark				
		Fail	3rd	2:2	2:1	1st
Introduction	10%	No introduction provided.	Limited introduction.	Moderate introduction addressing the key elements required.	Good, sharp introduction with suitable detail of all key elements.	Excellent introduction detailing all required components in a succinct manor.
Methodology	30%	No or limited description of the methodology.	Methodology provided with incomplete or missing information.	Primary components of the methodology presented to a good standard with some missing components.	All components of the methodology are addressed and described.	All components of the methodology are addressed, described and adequately justified where necessary.
Results	30%	No or limited results presented.	High level results presented with limited interpretation.	High-level results presented with some superficial interpretation.	Good selection of results shown in a clear and meaningful way with a fair level of interpretation.	Clear, succinct evaluation of the results with distinctive interpretation given to all aspects of differentiation.
Conclusions	20%	Limited or no conclusions drawn from results.	Limited, high-level conclusions drawn with limited links to the results section and some minor reflection.	Good range of conclusions drawn with some links to the results and an adequate study evaluation.	Good range of conclusions with clear supporting evidence from the results and an adequate evaluation of the study considering some future work.	Excellent range of conclusions drawn with distinct supporting evidence. The study has been evaluated and reflections are clear with relevant future work identified.
Presentation	10%	Poster is presented with limited structure, has substantial errors or uses text and images which are unreadable.	The poster has an evident structure but with unconsidered use of images, tables and text.	The poster has a good structure with appropriate use of images, tables and text.	The poster has a good structure with a good use of images, tables and text where appropriate. The poster has an evident but limited flow.	The poster has an exceptional structure with excellent use of images, text and tables. The flow of the poster is uninterrupted.